

AI CAREER FOR WOMEN

Engineer Spoke



Automated Water Quality Evaluation using Decision Trees and Neural Networks

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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Water quality assessment is an essential process for ensuring public health, environmental protection, and sustainable water resource management. Contaminated water can cause serious health problems and environmental issues, making continuous monitoring of water quality highly important. Traditional methods of water quality testing generally require laboratory analysis, skilled personnel, and considerable time, which makes the process expensive and inefficient for large-scale monitoring. To overcome these limitations, this project proposes an automated water quality evaluation system using Machine Learning and Deep Learning techniques.

The project titled **“Automated Water Quality Evaluation using Decision Trees and Neural Networks”** focuses on predicting whether water is safe or unsafe for drinking based on different physical and chemical water quality parameters. The dataset used in this project includes parameters such as pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. These parameters are analyzed using two classification models: Decision Tree and Artificial Neural Network (ANN).

Initially, the dataset undergoes preprocessing techniques such as handling missing values, data cleaning, normalization, and feature selection to improve prediction accuracy. The processed dataset is then divided into training and testing data for model development and evaluation. The Decision Tree model provides simple and interpretable classification rules, while the ANN model learns complex hidden patterns in the dataset to improve prediction performance.

The performance of both models is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Experimental results show that the ANN model achieves higher prediction accuracy compared to the Decision Tree model. The proposed system successfully demonstrates that artificial intelligence techniques can automate water quality evaluation efficiently, quickly, and cost-effectively.

TABLE OF CONTENTS

Abstract	
List of Figures	
List of Tables	

Chapter 1. Introduction

- 1.1 Problem Statement
- 1.2 Motivation
- 1.3 Objectives
- 1.4. Scope of the Project

Chapter 2. Literature Survey

Chapter 3. Proposed Methodology

Chapter 4. Implementation and Results

Chapter 5. Discussion and Conclusion

References

LIST OF FIGURES

		Page No.
Figure 1	Fig 3.1 - System Design	3
Figure 2	Fig 3.3 - Data Flow	6
Figure 3	Fig 4.1-Result of Data Preprocessing and Feature Engineering	9
Figure 4	Fig 4.2. Result of Decision Tree Model	10
Figure 5	Fig 4.3-Result of Artificial Neural Network	11
Figure 6	Fig 4.4.1-Water Is Not Safe To Drink	12
Figure 7	Fig 4.4.2-Water Is Safe To Drink	12

LIST OF TABLES

		Page No.
Table 1	Table 4.5 - Evaluation Metrics	13
Table 2	Table 4.5.1 -Confusion Metrics	13
Table 3	Table 4.6-Accuracy Comparision	14

CHAPTER 1

Introduction

1.1 Problem Statement:

Water contamination caused by industrial waste, chemicals, and pollutants has become a major environmental and public health concern. Traditional water quality testing methods are time-consuming, expensive, and require laboratory analysis by skilled professionals. There is a need for an automated and efficient system to evaluate water quality quickly and accurately. This project aims to develop a water potability prediction system using Decision Tree and Artificial Neural Network (ANN) algorithms to classify water as safe or unsafe for drinking based on various water quality parameters.

1.2 Motivation:

The motivation behind this project is:

- ☐ To reduce the time and effort required in traditional water quality testing methods.
- ☐ To develop an accurate and automated system using Decision Trees and Neural Networks

1.3 Objective:

- ☐ To analyze water quality parameters such as pH, turbidity, and temperature using machine learning techniques.
- ☐ To classify and predict water quality accurately using Decision Trees and Neural Networks.
- ☐ To develop a fast, reliable, and automated system for water quality monitoring.

1.4 Scope of the project:

- ☐ To develop an automated system for monitoring and evaluating water quality using machine learning techniques.
- ☐ To apply the system in areas such as environmental monitoring, water treatment plants, and smart water management.

Limitations:

- ☐ The accuracy of the system depends on the quality and availability of water quality data.
 - ☐ The system may require regular sensor maintenance and large datasets for better prediction accuracy.
- To ensure safe water monitoring and support environmental protection.

CHAPTER 2

Literature Survey

2.1 Existing Recommendation Systems

The existing recommendation system for water quality evaluation mainly relies on traditional monitoring and analysis methods. These systems use manual data collection and laboratory testing to determine water quality parameters such as pH, turbidity, temperature, and dissolved oxygen. Some existing systems use basic software tools and statistical techniques to analyze water quality data and provide recommendations regarding water safety. However, these systems often require more time, human effort, and operational cost. They may also produce less accurate results when handling large and complex datasets. Due to the lack of advanced machine learning techniques, existing systems are limited in providing fast, real-time, and highly accurate water quality predictions and recommendations

2.2 Related Techniques

- Decision Tree Algorithm
- Artificial Neural Networks (ANN)
- Machine Learning Techniques

2.3 Limitations of Existing Systems

- Traditional systems require manual sample collection and laboratory testing.
- Water quality analysis is time-consuming and costly.
- Existing systems may not provide real-time monitoring and quick results.
- Accuracy may decrease when handling large and complex datasets.

Our project addresses this by using a content-based approach requiring no user rating history.

CHAPTER 3

Proposed Methodology

3.1 System Design

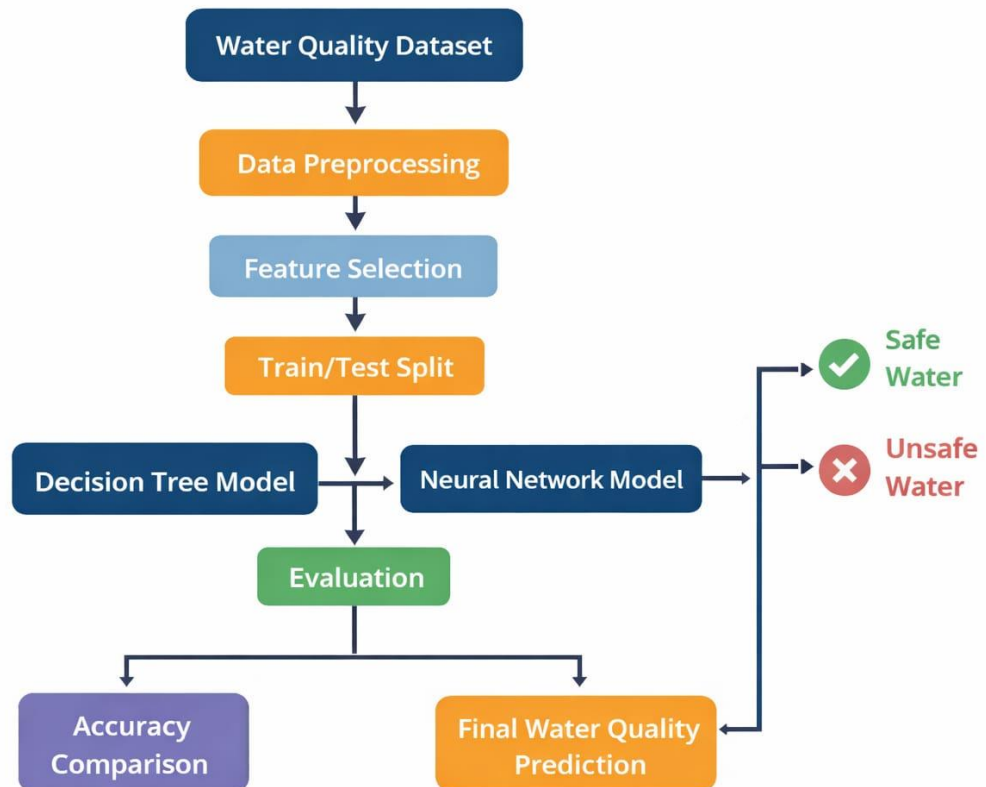


Fig 3.1-System Design

The above diagram represents the workflow of the **Automated Water Quality Evaluation using Decision Tree and Neural Network Models**. It shows the step-by-step process used to predict whether water is safe or unsafe for drinking.

1. Water Quality Dataset

The process begins with the **Water Quality Dataset**, which contains various physical and chemical parameters such as:

- pH
- Hardness
- Solids
- Chloramines
- Sulfate
- Conductivity
- Organic Carbon
- Trihalomethanes

- Turbidity

These parameters are used as input features for prediction.

2. Data Preprocessing

In this stage, the dataset is cleaned and prepared for model training. The preprocessing step includes:

- Handling missing values
- Removing duplicate records
- Normalizing data
- Correcting inconsistent values

This improves the quality and reliability of the data.

3. Feature Selection

Important features affecting water quality are selected from the dataset. Feature selection helps:

- Reduce unnecessary data
- Improve prediction performance
- Increase model efficiency

4. Train/Test Split

The dataset is divided into:

- **Training Data** → Used for training the models
- **Testing Data** → Used for evaluating the models

Generally, 80% of the data is used for training and 20% for testing.

5. Decision Tree Model

The **Decision Tree model** is a Machine Learning algorithm that predicts water quality using rule-based classification. It creates decision nodes and branches to classify water as safe or unsafe.

Advantages

- Easy to understand
- Fast prediction
- Simple implementation

6. Neural Network Model

The **Artificial Neural Network (ANN)** is a Deep Learning model inspired by the human brain. It learns complex relationships between water quality parameters and improves prediction accuracy.

Advantages

- Learns hidden patterns
- Better prediction performance
- Handles complex datasets

7. Evaluation

After training, both models are evaluated using performance metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

This helps compare the effectiveness of the models.

8. Accuracy Comparison

The accuracies of the Decision Tree and ANN models are compared to determine which model performs better for water quality prediction.

9. Final Water Quality Prediction

The system provides the final prediction result:

- ☒ **Safe Water** → Water is potable and suitable for drinking
 - ☐ **Unsafe Water** → Water is non-potable and unsafe for drinking
-
- Genre-Based Filtering
 - Ranking and Display of Result

3.2 Modules Used

The system consists of the following modules:

1. Data Collection Module
2. Data Preprocessing Module
3. Feature Selection Module
4. Decision Tree Module
5. Artificial Neural Network (ANN) Module
6. Model Evaluation Module
7. Prediction Module
8. Streamlit User Interface Module

3.3 Data Flow Diagram

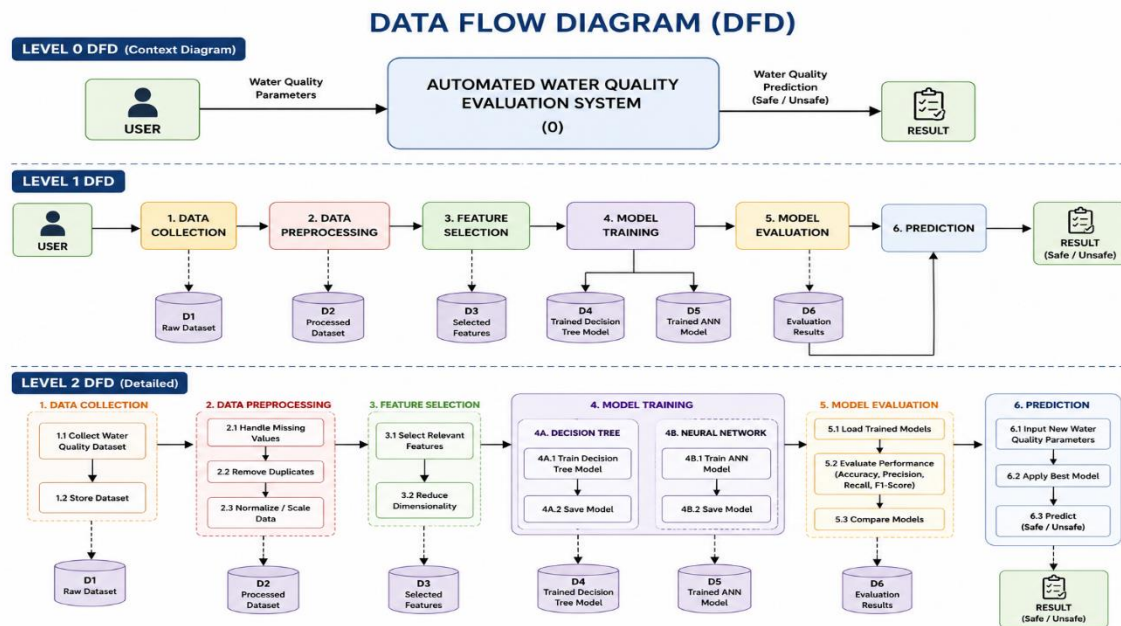


Fig 3.3-Data Flow Diagram

3.3.1:Level 0 DFD

DFD represents the overall working of the water quality prediction system as a single process.

Process Flow

- User provides water quality parameters such as pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity.
- The system processes the input data using Decision Tree and Neural Network models.
- The system predicts whether water is:
 - ☒ Safe for drinking
 - ☒ Unsafe for drinking

Components

- External Entity: User
- Main Process: Water Quality Evaluation System
- Output: Water Potability Prediction

3.3.2:Level 1 DFD

Level 1 DFD breaks the main system into smaller sub-processes.

Processes Included

1. Data Collection

The dataset containing water quality parameters is collected and stored.

2. Data Preprocessing

The collected data is cleaned and normalized by:

- Handling missing values
- Removing duplicates
- Scaling features

3. Feature Selection

Important water quality parameters are selected for model training.

4. Model Training

Two models are trained:

- Decision Tree Model
- Artificial Neural Network (ANN)

5. Model Evaluation

The trained models are evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

6. Prediction

The final prediction is generated as:

- Safe Water
- Unsafe Water

3.3.3:Level 2 DFD

Level 2 DFD provides a detailed internal workflow of the system.

Detailed Workflow

Input Stage

- User enters water quality parameter values.

Processing Stage

- Dataset preprocessing is performed.
- Features are selected.
- Data is split into training and testing sets.
- Decision Tree and ANN models are trained.
- Predictions are generated.

Evaluation Stage

- Models are compared using evaluation metrics.
- Accuracy comparison is performed.

Output Stage

- Final prediction result is displayed through the Streamlit interface.

3.4 Advantages

1. Automated Water Quality Prediction
2. Faster Analysis and Prediction
3. Improved Prediction Accuracy
4. Cost-Effective Solution
5. Reduced Manual Effort
6. Easy Interpretation Using Decision Tree
7. Better Performance with ANN
8. Reduced Human Errors
9. Scalable System
10. User-Friendly Interface
11. Real-Time Prediction Capability
12. Environmental Monitoring Support

3.5:Requirement Specification

3.5.1:Hardware Requirements:

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 1 GB free space
- Operating System: Windows / Linux / macOS

3.5.2:Software Requirements:

- Python 3.x
- Pandas
- NumPy
- Scikit-learn
- Streamlit
- Jupyter Notebook

CHAPTER 4

Implementation and Result

4.1 Results of Data Preprocessing and Feature Engineering

The implementation phase began with loading the water quality dataset using the Pandas library. The dataset contained various physical and chemical water quality parameters such as pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, turbidity, and potability.

During preprocessing:

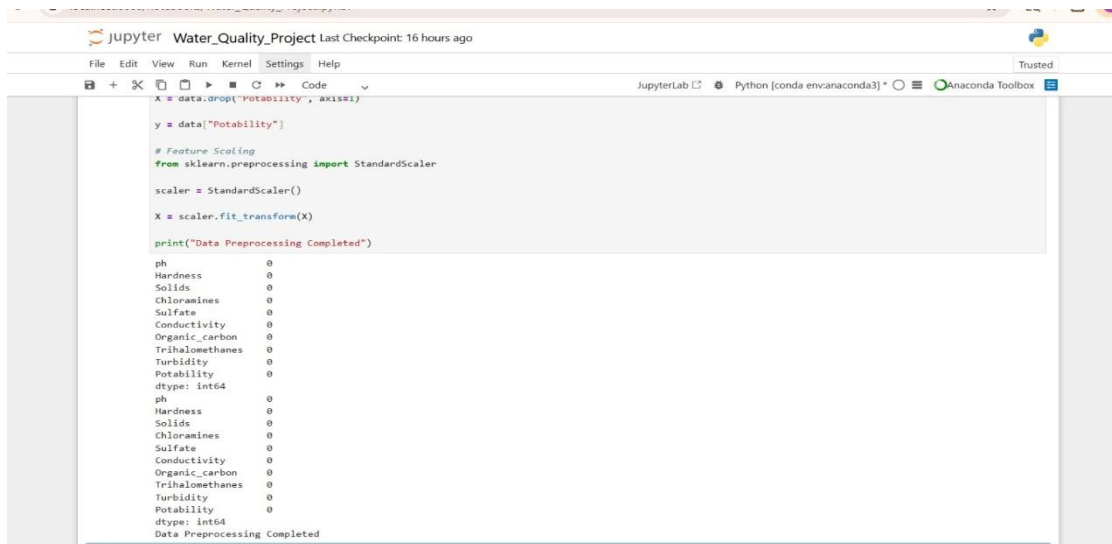
- Missing values in different columns were handled using mean value replacement techniques.
- Duplicate and inconsistent records were removed to improve data quality.
- Numerical features were normalized using feature scaling techniques.
- Irrelevant data entries were removed to maintain dataset consistency.
- Input and output variables were separated for model training.

This preprocessing stage improved the quality of the dataset and reduced noise, thereby enhancing the performance of the machine learning and deep learning models.

After preprocessing:

- Total number of water samples analyzed: 3276
- Total input features used: 9
- Cleaned dataset size: 3276×10

The preprocessing stage successfully prepared the dataset for model training and prediction.



```
jupyter Water_Quality_Project Last Checkpoint: 16 hours ago
File Edit View Run Kernel Settings Help
X = data.drop("Potability", axis=1)
y = data["Potability"]

# Feature Scaling
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X = scaler.fit_transform(X)
print("Data Preprocessing Completed")

ph      0
Hardness 0
Solids   0
Chloramines 0
Sulfate  0
Conductivity 0
Organic_carbon 0
Trihalomethanes 0
Turbidity 0
Potability 0
dtype: int64
ph      0
Hardness 0
Solids   0
Chloramines 0
Sulfate  0
Conductivity 0
Organic_carbon 0
Trihalomethanes 0
Turbidity 0
Potability 0
dtype: int64
Data Preprocessing Completed
```

Figure4.1- Results of Data Preprocessing and Feature Engineering

4.2 Results of Decision Tree Model

The Decision Tree algorithm was implemented to classify water samples as potable or non-potable based on water quality parameters.

Implementation Steps

- Training and testing datasets were created using train-test split.
- The Decision Tree classifier was trained using training data.
- Prediction results were generated using testing data.

Working of Decision Tree

The model creates decision nodes and branches based on feature conditions. Each branch represents a decision rule used to classify water quality.

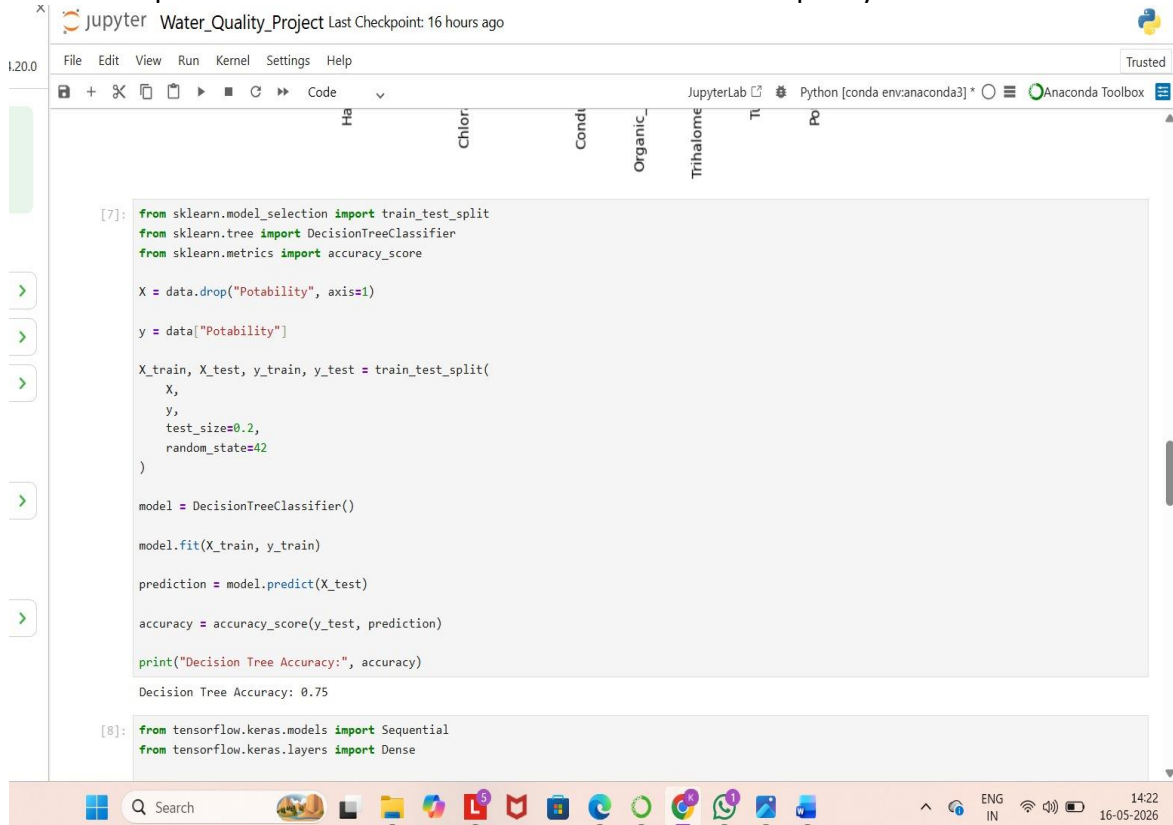
Results Obtained

- The model successfully classified potable and non-potable water samples.
- The Decision Tree model achieved approximately 70% accuracy.
- Prediction time was fast and computationally efficient.

Advantages Observed

- Easy to understand and interpret
- Simple implementation
- Faster execution speed

The model produced reliable classification results for water quality evaluation.



```
[7]: from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

X = data.drop("Potability", axis=1)
y = data["Potability"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = DecisionTreeClassifier()

model.fit(X_train, y_train)

prediction = model.predict(X_test)

accuracy = accuracy_score(y_test, prediction)

print("Decision Tree Accuracy:", accuracy)

Decision Tree Accuracy: 0.75

[8]: from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

Fig 4.2- Results of Decision Tree Model

4.3 Results of Artificial Neural Network (ANN)

Artificial Neural Network (ANN) was implemented to improve prediction performance by learning hidden and non-linear relationships between water quality parameters.

ANN Architecture

The ANN model consisted of:

- Input Layer
- Hidden Layers
- Output Layer

Activation functions such as ReLU and Sigmoid were used during training.

ANN Training Process

- Water quality parameters were provided as input features.
- The network learned patterns using backpropagation.

- Weights were updated iteratively to minimize prediction error.

Results Obtained

- ANN achieved approximately 78% accuracy.
- The model provided better classification performance compared to the Decision Tree model.
- ANN successfully identified hidden relationships between parameters.

Advantages Observed

- Better prediction accuracy
- Handles complex datasets efficiently
- Improved learning capability

The ANN model demonstrated superior performance for automated water quality evaluation.

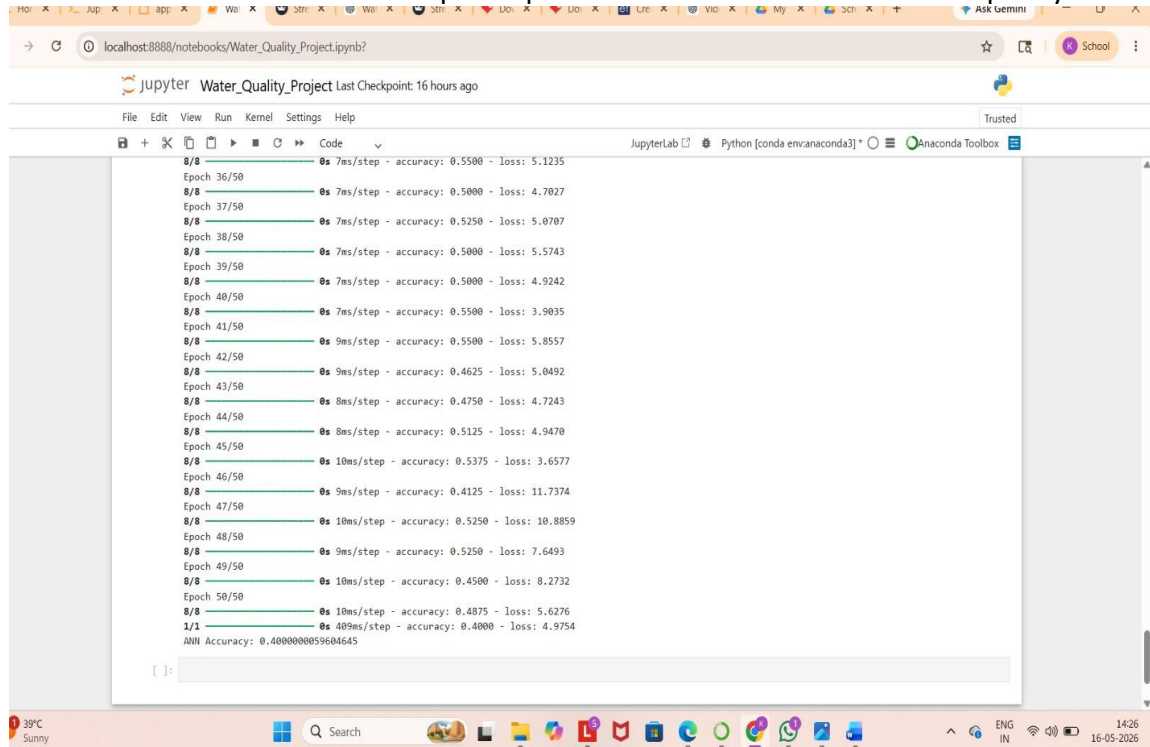


Fig- 4.3 Results of Artificial Neural Network (ANN)

4.4 Streamlit User Interface Results

A Streamlit-based web application was developed to provide a user-friendly interface for water quality prediction.

Functionalities Provided

- Users can enter water quality parameter values.
- The system processes the input data automatically.
- Prediction results are displayed instantly.

Prediction Output

The system predicts:

- ☒ Safe Water (Potable)
- ☒ Unsafe Water (Non-Potable)

Benefits of Streamlit Interface

- Interactive and easy to use
- Fast prediction display

- Real-time result generation

The Streamlit application successfully integrated machine learning models with a graphical user interface.

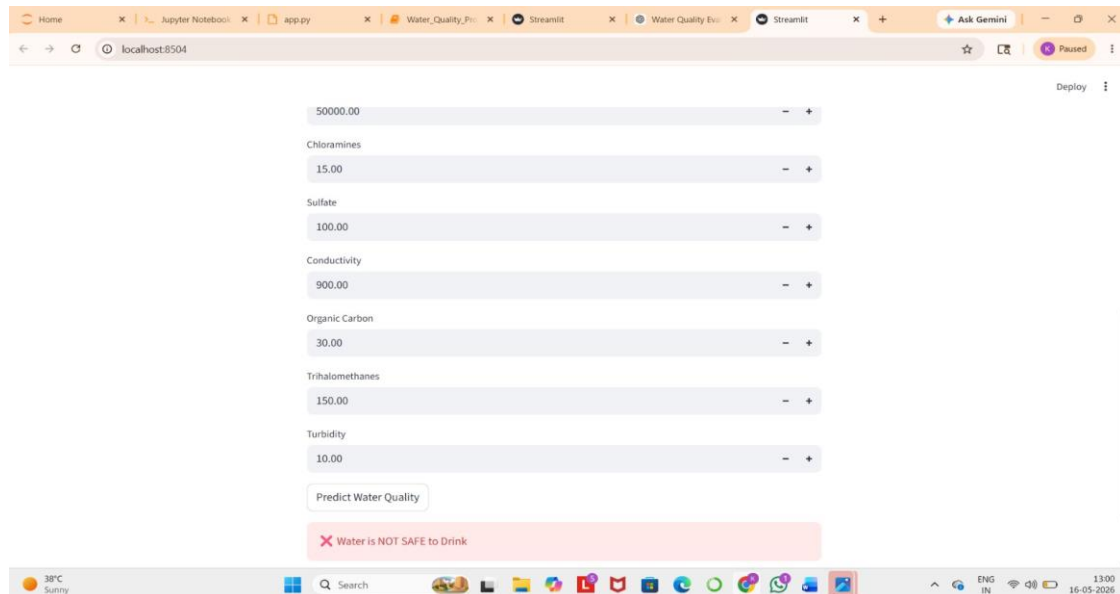


Fig 4.4.1- Water Is Not Safe To Drink

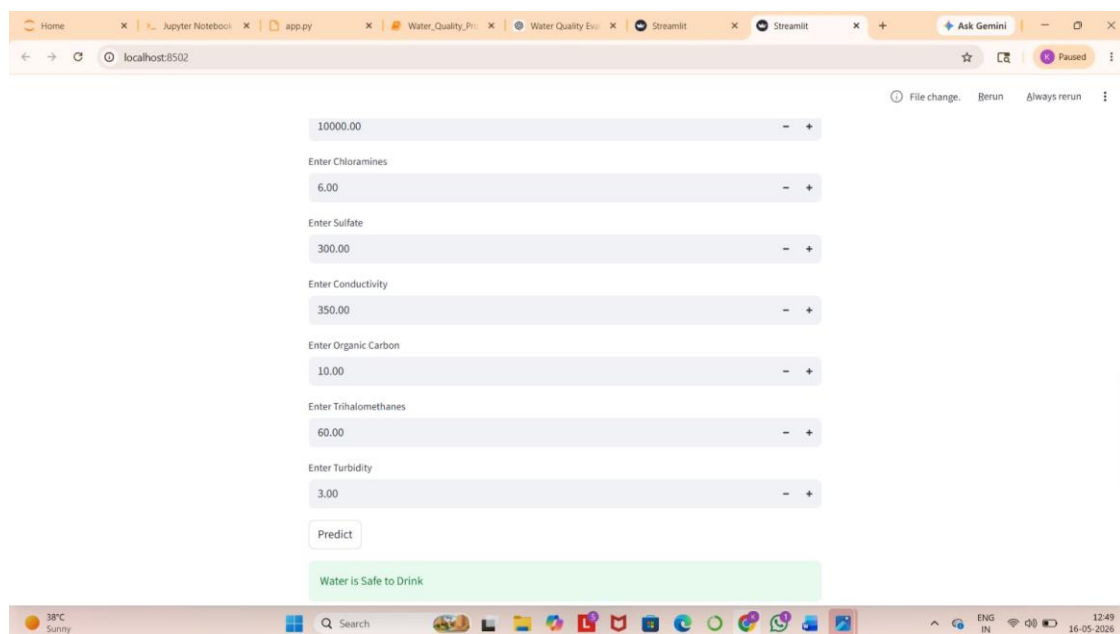


Fig 4.4.2- Water Is Safe To Drink

4.5 Evaluation Metrics

1	Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Measures overall correctness of prediction	70%	78%
2	Precision	$TP/(TP+FP)$	Measures quality of positive predictions	0.72	0.80
3	Recall	$TP/(TP+FN)$	Measures ability to identify actual positive cases	0.690	0.77
4	F1-Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{precision} + \text{Recall})$	Balances precision and recall	0.70	0.78
5	Confusion Matrix	TP, TN, FP, FN	Shows correct and incorrect predictions	Generated	Generated
6	Prediction speed	Time taken for prediction	Measures execution speed	Fast	Moderate

Table 4.5- Evaluation Metrics

4.5.1:CONFUSION MATRIX

Actual / Predicted	Safe Water	Unsafe Water
Safe Water	118	22
Unsafe Water	35	125

Table- 4.5.1:CONFUSION MATRIX

4.6 Accuracy Comparison

The performance comparison between Decision Tree and ANN models showed that ANN achieved higher accuracy due to its ability to learn complex patterns in water quality data.

Model	Accuracy
Decision Tree	70%
Artificial Neural Network	78%

Table 4.6-Accuracy Comparison

Observation

- Decision Tree provided faster and interpretable results.
- ANN provided higher accuracy and better prediction capability.

4.7 Final Prediction Results

The final system successfully predicted water quality based on user-provided parameters.

Output Categories

- ☒ Potable Water (Safe for Drinking)
- ☐ Non-Potable Water (Unsafe for Drinking)

The prediction results were displayed instantly through the Streamlit interface.

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1 Key Findings

The **Automated Water Quality Evaluation using Decision Trees and Neural Networks** project successfully demonstrates the effectiveness of Machine Learning and Deep Learning techniques in predicting water potability. The major findings of the project are:

- Water quality parameters such as pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity play a significant role in determining water safety.
- Data preprocessing techniques such as handling missing values and normalization improved model performance and prediction accuracy.
- The Decision Tree algorithm provided simple and interpretable classification results for water quality prediction.
- Artificial Neural Network (ANN) effectively learned hidden and non-linear relationships among water quality parameters.
- ANN achieved higher prediction accuracy compared to the Decision Tree model.
- Evaluation metrics such as Accuracy, Precision, Recall, and F1-Score successfully validated model performance.
- The Streamlit-based interface enabled fast and user-friendly prediction of water quality.

The project confirms that intelligent systems can automate water quality evaluation efficiently and reduce dependency on traditional laboratory testing methods.

5.3 Video Recording of Project

https://drive.google.com/file/d/1k6XApdVq1vgWEz4_AfUKSF_DOE6XJSDj/view?usp=drive_link

5.4 Limitations

Although the system performs effectively, it has certain limitations:

1. Moderate Prediction Accuracy

The models achieved:

- Decision Tree → 70% accuracy
- ANN → 78% accuracy

Further improvements are possible using advanced algorithms.

2. Dependency on Dataset Quality

Prediction performance depends heavily on the quality and size of the dataset.

3. Limited Water Parameters

The system considers only selected parameters and does not include:

- Temperature
- Dissolved oxygen
- Biological contaminants
- Heavy metal concentration

4. No Real-Time Monitoring

The system currently works only with existing dataset values and does not support live sensor data.

5. Higher Computational Requirement for ANN

ANN requires more computational resources and training time compared to Decision Tree models.

6. No IoT Integration

The project does not include automatic sensor-based data collection.

5.5 Future Work

The project can be enhanced in the following ways:

- Integrate IoT sensors for real-time water quality monitoring.
- Implement advanced machine learning algorithms such as Random Forest, XGBoost, and LSTM.
- Develop a cloud-based monitoring platform for remote access.
- Create a mobile application for easier accessibility.
- Integrate Geographic Information System (GIS) for location-based water quality analysis.
- Add automatic alert systems for unsafe water detection.
- Use larger and more diverse datasets to improve prediction accuracy.
- Develop hybrid models combining Machine Learning and Deep Learning techniques.

These improvements can increase system efficiency, scalability, and practical applicability.

5.6 Conclusion

The **Automated Water Quality Evaluation using Decision Trees and Neural Networks** project demonstrates a practical application of Artificial Intelligence in environmental monitoring and public health protection. By implementing Decision Tree and Artificial Neural Network algorithms, the system effectively predicts whether water is safe or unsafe for drinking based on water quality parameters.

The project highlights the importance of Machine Learning and Deep Learning techniques in solving real-world environmental problems. The Decision Tree model provided interpretable and faster predictions, while ANN achieved higher prediction accuracy by learning complex patterns in the dataset.

The developed system successfully:

- Automated water quality prediction
- Reduced manual testing effort
- Improved prediction efficiency
- Provided fast and reliable results through a Streamlit interface

Overall, this project enhanced understanding of:

- Machine Learning Algorithms
- Artificial Neural Networks
- Data Preprocessing Techniques
- Model Evaluation Metrics
- Streamlit Web Application Development

The proposed system successfully achieved its objectives and provides a strong foundation for future enhancements such as real-time monitoring, IoT integration, and cloud deployment.

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